

St George Girls High School

Year 11 – Higher School Certificate Course

Assessment Task 1

December 2007



Mathematics

Extension 1

*Time Allowed: 75 Minutes
(plus 5 minutes reading time)*

Instructions to Candidates

1. Write using black or blue pen.
2. Attempt all questions.
3. Start each question on a new page.
4. Show all necessary working.
5. Marks for each question are shown in the right column.
6. Complete cover sheet clearly showing:
 - your name
 - your mathematics class and teacher.

Question 1 – (12 marks) – Start a New Page

Marks

- a) Find the centre and radius of the circle with equation given by:

$$x^2 - 8x + y^2 + 14y + 1 = 0$$

3

- b) (i) Derive the equation of the normal to the parabola $x^2 = 4ay$ at the point $(2at, at^2)$

3

- (ii) Find the coordinates of the point N where this normal meets the axis of the parabola.

- c) Given that $f(x) = 3^x$ complete the table.

3

x	0	1	2	3	4
3^x					

With these five function values estimate $\int_0^4 3^x dx$ using Simpson's Rule.

- d) (i) Differentiate $\sqrt{1+x^4}$

3

- (ii) Hence find $\int \frac{x^3 dx}{\sqrt{1+x^4}}$

Question 2 – (12 marks) – Start a new page

Marks

- a) For the function $y = 4x^3 - x^4$ 8
- (i) Show that there are 2 stationary points.
 - (ii) Find the nature of the stationary points.
 - (iii) Find the point(s) on the curve where the concavity changes.
 - (iv) Sketch the curve showing intercepts and the above features.
- b) Find the equation of the parabola with axis parallel to the y-axis and passing through the points $(0, -2)$, $(1, 0)$ and $(3, -8)$ 4

Question 3 – (12 marks) – Start a new page

Marks

- a) A parabola has its vertex at the point $(2, 1)$ and focus at the point $(2, 3)$.

3

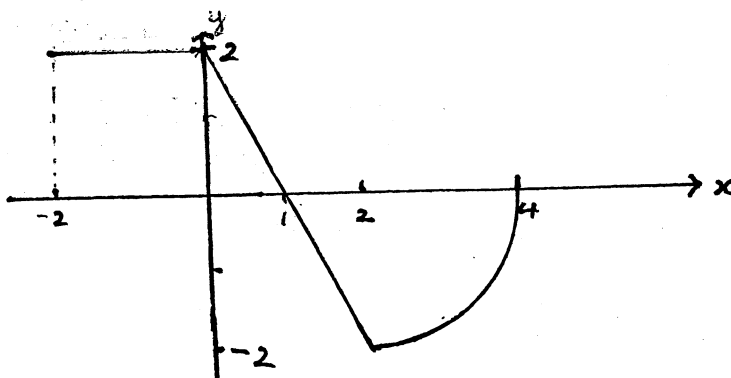
- (i) What is the focal length?
- (ii) What is the equation of the directrix?
- (iii) What is the equation of the parabola?

- b) Derive the equation of the locus of a point P that moves so that PA is twice the distance PB where A is $(0, 3)$ and B is $(4, 7)$.

3

- c) Use area formulae to evaluate $\int_{-2}^4 f(x) dx$

2



4

- d) Evaluate:

(i) $\int_1^3 (5 - 4x) dx$

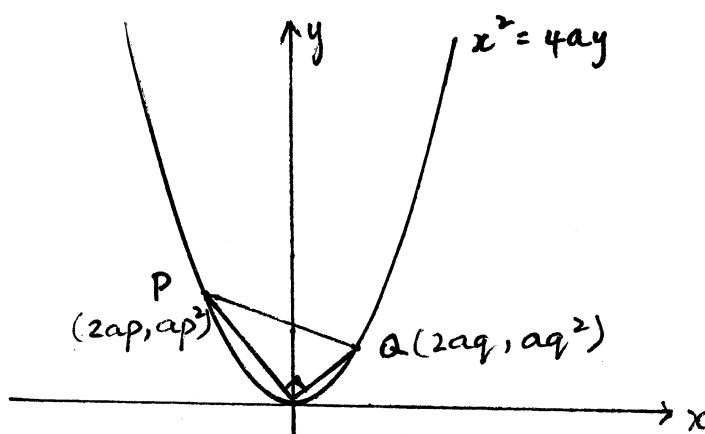
(iii) $\int_0^2 (2x - 1)^3 dx$

Question 4 – (12 marks) – Start a new page

Marks

- a) For the function $y = f(x)$, $f(-1) = 3$ and $f(3) = 1$. Sketch the function from $x = -1$ to $x = 3$ if over this domain $f'(x) < 0$ and $f''(x) < 0$, and the function is continuous. 3
- b) A curve contains the point $(1, 4)$ and its gradient is given by the function $(x+1)(x-2)$. Find the equation of the curve. 3

c)



The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two variable points on the parabola $x^2 = 4ay$. The chord PQ subtends a right angle at the origin. 6

(i) Find the gradient of OP

(ii) Show that $pq = -4$

M is the midpoint of PQ

(iii) Find the equation of the locus of M as P and Q vary.

(iv) Describe this locus.

Question 5 – (12 marks) – Start a new page

Marks

a) (i) On the same axes sketch the graphs of $y = \sqrt{x}$ and $y = 6 - x$.

6

(ii) Show that $(4, 2)$ is the point of intersection of the two graphs.

(iii) Find the area bounded by the two graphs and the y-axis.

b) A rectangle is to be inscribed in the semicircle $y = \sqrt{4 - x^2}$

6

(i) Show that the area of the rectangle can be given by $2x\sqrt{4 - x^2}$

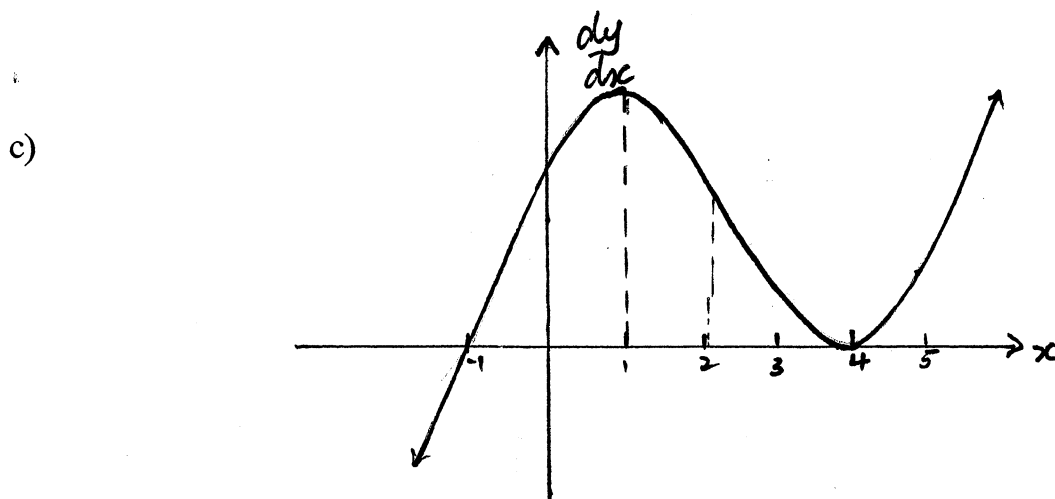
(ii) Find the dimensions of the rectangle with the greatest area.

Question 6 – (12 marks) – Start a new page

Marks

a) Show that $\int x(x+3) dx \neq \int x dx \int (x+3) dx$ 2

b) Find the volume of the solid formed when the region bounded by the parabola $y = 4 - x^2$ and the x -axis is rotated about the x -axis. 4



Given the graph of the derivative of y , state values of the domain where the function y : 6

- (i) is increasing
- (ii) has a minimum turning point
- (iii) has inflexional points
- (iv) is concave up

QUESTION 1

a) $x^2 - 8x + y^2 + 14y + 1 = 0$

$$x^2 - 8x + 16 + y^2 + 14y + 49 = -1 + 16 + 49$$

$$(x-4)^2 + (y+7)^2 = 64$$

$$= 8^2$$

∴ THE CENTRE IS AT $(4, -7)$; THE RADIUS IS 8.

b) (i) $x^2 = 4ay$

$$\therefore y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$$

When $x = 2at$, $\frac{dy}{dx} = t$

Gradient of normal $= -\frac{1}{t}$

∴ Equation of normal is

$$y - at^2 = -\frac{1}{t}(x - 2at)$$

$$ty - at^3 = -x + 2at$$

$$x + ty - at^3 - 2at = 0$$

(ii) Let $x = 0$

$$ty - at^3 - 2at = 0$$

$$y = at^2 + 2a$$

∴ N is the point $(0, at^2 + 2a)$

c) $f(x) = 3^x$

x	0	1	2	3	4
3^x	1	3	9	27	81

$$\int_a^b f(x) dx \doteq \frac{b-a}{6} \left\{ f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right\}$$

$$\begin{aligned} \therefore \int_0^4 3^x dx &\doteq \frac{2}{6} \{ 1 + 4 \times 3 + 2 \times 9 + 4 \times 27 + 81 \} \\ &= 73\frac{1}{3}. \end{aligned}$$

$$\begin{aligned} d) \text{ (i) } \frac{d}{dx} \sqrt{1+x^4} &= \frac{d}{dx} (1+x^4)^{\frac{1}{2}} \\ &= \frac{1}{2} (1+x^4)^{-\frac{1}{2}} \cdot 4x^3 \\ &= \frac{2x^3}{\sqrt{1+x^4}} \end{aligned}$$

$$\begin{aligned} \text{(ii')} \int \frac{x^3}{\sqrt{1+x^4}} dx &= \frac{1}{2} \int \frac{2x^3}{\sqrt{1+x^4}} dx \\ &= \frac{1}{2} \sqrt{1+x^4} + C \end{aligned}$$

QUESTION 2

a) $y = 4x^3 - x^4$

(i) $\frac{dy}{dx} = 12x^2 - 4x^3$

Let $\frac{dy}{dx} = 0$, $12x^2 - 4x^3 = 0$
 $4x^2(3 - x) = 0$

$\therefore x = 0$ or $x = 3$

When $x = 0$, $y = 0$

$x = 3$, $y = 4 \times 27 - 81$
 $= 27$

$\therefore$ There are 2 stationary points, $(0, 0)$ and $(3, 27)$

(ii) $\frac{d^2y}{dx^2} = 24x - 12x^2$

When $x = 3$, $\frac{d^2y}{dx^2} = 72 - 108$
 $\frac{d^2y}{dx^2} < 0$

$\therefore$ there is a maximum turning point at $(3, 27)$

When $x = 0$, $\frac{d^2y}{dx^2} = 0$

x	-1	0	1
$\frac{d^2y}{dx^2}$	-36	0	12

From the table, the concavity changes at $(0, 0)$

Hence there is a horizontal point of inflection at $(0, 0)$.

(iii) Let $\frac{d^2y}{dx^2} = 0$, $24x - 12x^2 = 0$
 $12x(2 - x) = 0$

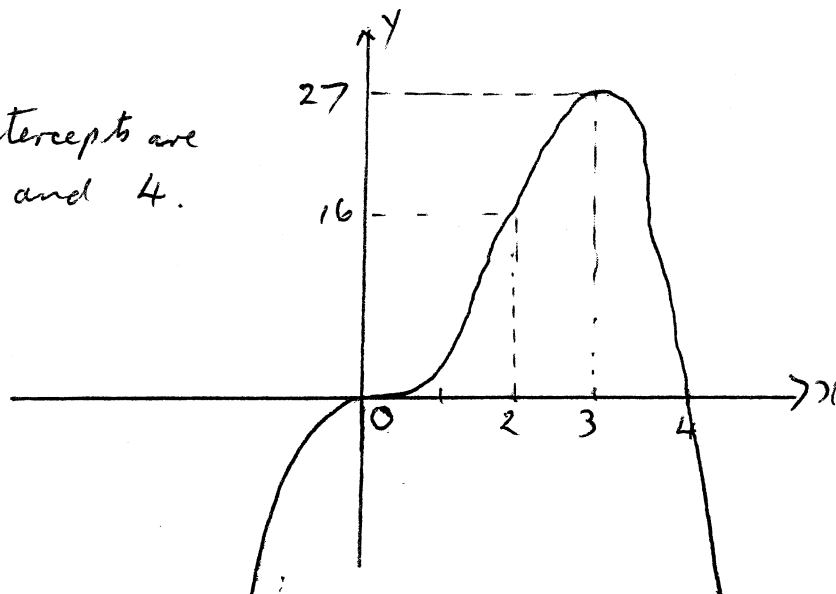
$x = 0$ or $x = 2$

When $x = 2$, $y = 4 \times 8 - 16$
 $= 16$

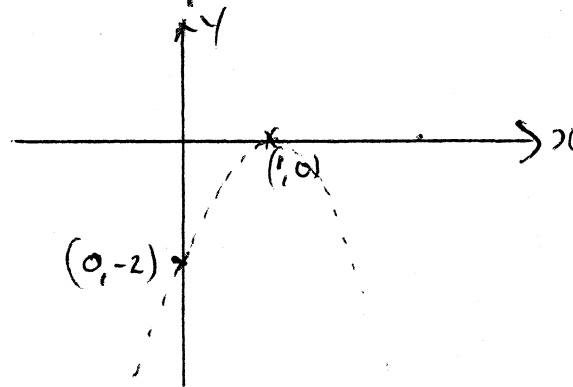
$\therefore$ Concavity changes at $(0, 0)$ and $(2, 16)$

(iv)

x intercepts are
0 and 4.



b)



(3, -8)

$$y = ax^2 + bx + c$$

Substitute $x=0, y=-2$

$$-2 = 0 + 0 + c$$

$$\therefore y = ax^2 + bx - 2$$

Subst. $x=1, y=0$

$$0 = a + b - 2$$

①

Subst $x=3, y=-8$

$$-8 = 9a + 3b - 2$$

$$0 = 9a + 3b + 6$$

$$0 = 3a + b + 2$$

②

$$\textcircled{2} - \textcircled{1}$$

$$2a + 4 = 0$$

$$a = -2$$

Subst in ①

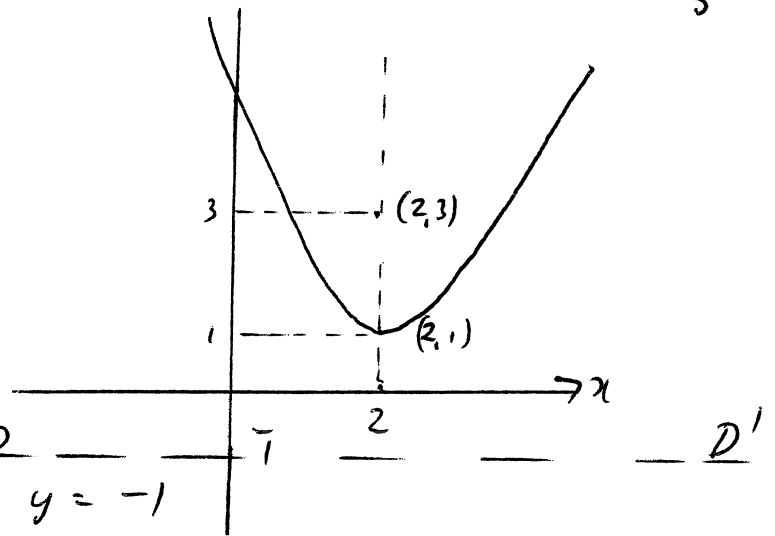
$$b = 4$$

$\therefore$ Equation is
 $y = -2x^2 + 4x - 2$

QUESTION 3

a)

(i) Focal length:
 $a = 2$



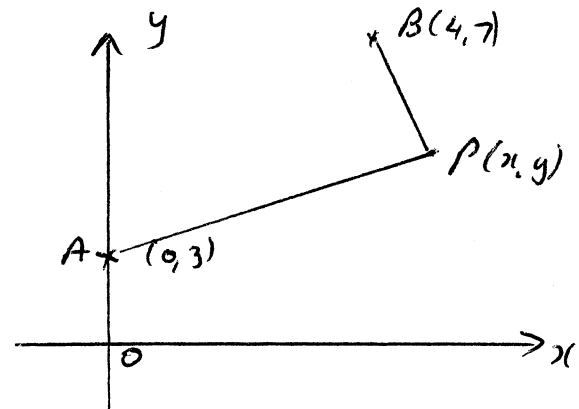
(ii) Directrix is $y = -1$

(iii) Equation is $(x-2)^2 = 4 \times 2 (y-1)$
 $(x-2)^2 = 8(y-1)$

b) Let P be the point (x, y)

$$PA = 2 PB$$

$$PA^2 = 4 PB^2$$



$$x^2 + (y-3)^2 = 4[(x-4)^2 + (y-7)^2]$$

$$x^2 + y^2 - 6y + 9 = 4[x^2 - 8x + 16 + y^2 - 14y + 49]$$

$$x^2 + y^2 - 6y + 9 = 4x^2 - 32x + 4y^2 - 56y + 260$$

$$\therefore 3x^2 + 3y^2 - 32x - 50y + 251 = 0$$

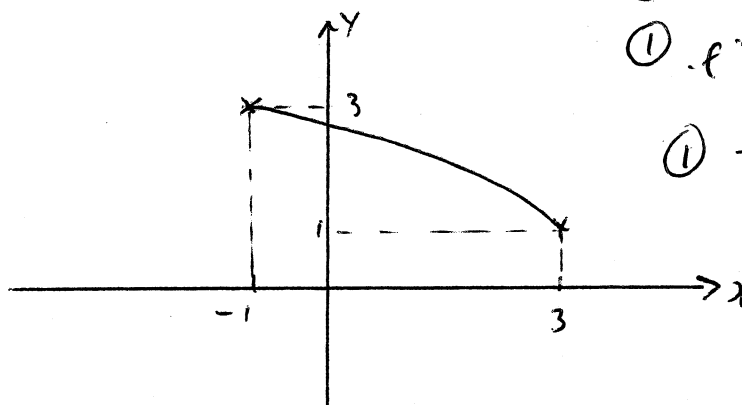
$$\begin{aligned} \text{c) } \int_{-2}^4 f(x) &= \frac{1}{2} \times 1 \times 2 - \frac{1}{2} \times 1 \times 2 - \frac{1}{4} \times \pi \times 2^2 + 4 \\ &= 4 - \pi \end{aligned}$$

$$\begin{aligned} \text{d) (i) } \int_1^3 (5-4x) dx &= [5x - 2x^2]_1^3 \\ &= (15 - 18) - (5 - 2) \\ &= -6 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad \int_0^2 (2x-1)^3 dx &= \left[\frac{(2x-1)^4}{4 \times 2} \right]_0^2 \\
 &= \frac{1}{8} (3^4 - (-1)^4) \\
 &= \frac{1}{8} \times 80 \\
 &= 10
 \end{aligned}$$

QUESTION 4

a)

①. $f'(x) < 0$, decreasing①. $f''(x) < 0$, concave down

① for scale, and pts.

$$\begin{aligned}
 \text{b)} \quad \frac{dy}{dx} &= (x+1)(x-2) \\
 &= x^2 - x - 2
 \end{aligned}$$

 $\frac{1}{2}$

1

$$\therefore y = \frac{x^3}{3} - \frac{x^2}{2} - 2x + C$$

$$\text{Subst } x=1, y=4$$

1

$$4 = \frac{1}{3} - \frac{1}{2} - 2 + C$$

$$\therefore C = \frac{37}{6}$$

 $\frac{1}{2}$

$$\therefore y = \frac{x^3}{3} - \frac{x^2}{2} - 2x + \frac{37}{6}$$

Chad

$$6y = 2x^3 - 3x^2 - 12x + 37$$

$$\begin{aligned} \text{c) (i)} \quad M_{OP} &= \frac{ap^2}{2ap} \\ &= \frac{p}{2} \end{aligned} \quad - (1)$$

$$\text{(ii)} \quad M_{OQ} = \frac{q}{2}$$

$$PO \perp OQ.$$

$$\therefore \frac{p}{2} \cdot \frac{q}{2} = -1 \quad - (1)$$

$$pq = -4$$

$$\begin{aligned} \text{(iii)} \quad \text{MIDPOINT OF } PQ \text{ IS } &\left(\frac{2ap+2aq}{2}, \frac{ap^2+aq^2}{2} \right) \\ &= \left(a(p+q), \frac{a(p^2+q^2)}{2} \right) - (1) \end{aligned}$$

$\therefore$ Parametric equations of M are

$$x = a(p+q) \quad (1)$$

$$y = \frac{a(p^2+q^2)}{2}$$

$$y = \frac{a}{2} ((p+q)^2 - 2pq) \quad (2)$$

$$\text{From (1) } p+q = \frac{x}{a}, \text{ and } pq = -4 \quad - (1)$$

Sub in (2)

$$y = \frac{a}{2} \left(\left(\frac{x}{a} \right)^2 - 2(-4) \right)$$

$$y = \frac{a}{2} \left(\frac{x^2}{a^2} + 8 \right)$$

$$2y = \frac{x^2}{a} + 8a \quad - (1)$$

$$2ay = x^2 + 8a^2$$

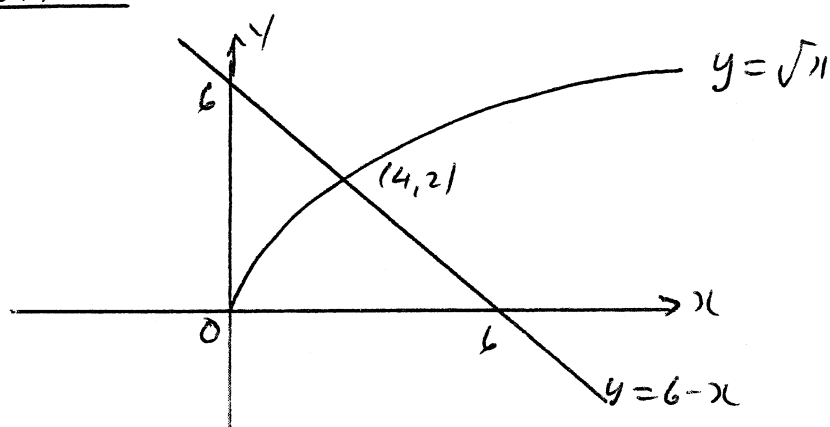
$$\therefore x^2 = 2ay - 8a^2$$

$$x^2 = 2a(y - 4a)$$

(iv) $\therefore$ Locus is a parabola with vertex at $(0, 4a)$ with focal length $\frac{a}{2}$ - (1)

QUESTION 5

a) (i)



$$(i) \quad y = \sqrt{x}$$

$$y = 6 - x$$

$$\therefore \sqrt{x} = 6 - x$$

$$x = (6 - x)^2$$

$$= 36 - 12x + x^2$$

$$\text{i.e. } x^2 - 13x + 36 = 0$$

$$(x - 4)(x - 9) = 0$$

$$x = 4 \text{ or } x = 9$$

$$\text{When } x = 4, \quad y = 6 - 4$$

$$= 2$$

$\therefore (4, 2)$ is the point of intersection of the 2 graphs

[Note: when $x = 9$, $y = -3$ which is not on the graph $y = \sqrt{x}$.]

$$(iii) \quad A = \int_0^4 x^{\frac{1}{2}} dx + \int_4^6 (6 - x) dx$$

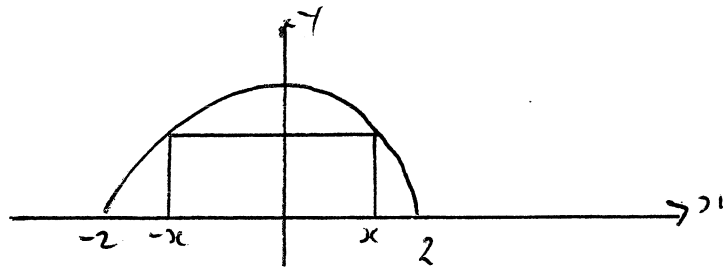
$$= \left[\frac{2}{3} x^{\frac{3}{2}} \right]_0^4 + \left[6x - \frac{x^2}{2} \right]_4^6$$

$$= \frac{16}{3} - 0 + (36 - 18) - (24 - 8)$$

$$= \frac{16}{3} + 18 - 16$$

$$= 7\frac{1}{3}$$

6)



(i) Let one side of the rectangle lie on the x axis between $-x$ and x (as symmetry)

$$\therefore \text{Length} = 2x$$

$$\text{Height} = \sqrt{4-x^2}$$

$$\therefore \text{Area} = 2x \sqrt{4-x^2}$$

(ii) Let $A = 2x(4-x^2)^{\frac{1}{2}}$

$$\frac{dA}{dx} = 2x \cdot \frac{1}{2} (4-x^2)^{-\frac{1}{2}} \cdot (-2x) + (4-x^2)^{\frac{1}{2}} \cdot 2$$

$$= -\frac{2x^2}{\sqrt{4-x^2}} + 2\sqrt{4-x^2}$$

$$= \frac{-2x^2 + 2(4-x^2)}{\sqrt{4-x^2}}$$

$$= \frac{8-4x^2}{\sqrt{4-x^2}}$$

$$\text{Let } \frac{dA}{dx} = 0 \quad \text{ie } \frac{8-4x^2}{\sqrt{4-x^2}} = 0$$

$$8-4x^2 = 0$$

$$2-x^2 = 0$$

$$x = \pm \sqrt{2}$$

x	1	$\sqrt{2}$	1.5
$\frac{dA}{dx}$	+	0	-

$\therefore$ Area will be a max

when $x = \sqrt{2}$

$$\text{When } x = \sqrt{2}, \quad y = \sqrt{4-(\sqrt{2})^2} = \sqrt{2}$$

$\therefore$ Dimensions are $2\sqrt{2} \times 2$.

QUESTION 6

$$a) \quad \int x(x+3) dx = \int x^2 + 3x dx$$

$$= \frac{x^3}{3} + \frac{3x^2}{2} + C$$

$$\int x dx \cdot \int (x+3) dx = \frac{x^2}{2} \left(\frac{x^2}{2} + 3x \right) + C$$

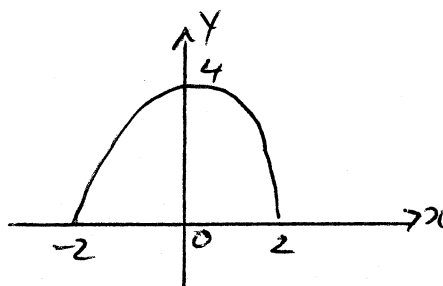
$$= \frac{x^4}{4} + \frac{3x^3}{2} + C$$

$$\neq \int x(x+3) dx.$$

b)

$$V = \pi \int_a^b [f(x)]^2 dx$$

$$= \pi \int_{-2}^2 (4-x^2)^2 dx$$



$$= 2\pi \int_0^2 (16 - 8x^2 + x^4) dx$$

$$= 2\pi \left[16x - \frac{8x^3}{3} + \frac{x^5}{5} \right]_0^2$$

$$= 2\pi \left[32 - \frac{64}{3} + \frac{32}{5} - 0 \right]$$

$$= \frac{512}{15} \pi$$

- c) (i) $x > -1$, $x \neq 4$
 (ii) $x = -1$
 (iii) $x = 1, 4$
 (iv) $x < -1$